Pharmacoeconomics and Safe Medication Use

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Learning Objectives

1. Apply the principles of pharmacoeconomics to patient care.
2. Differentiate between a medication error, an adverse drug event (ADE), a preventable ADE, and an adverse drug reaction (ADR).
3. Design an ADE reporting program, including committee structure, committee reporting mechanisms, and methods of detecting, reporting, and managing ADEs.
4. Provide recommendations for improving medication use safety using the 2017 SCCM Safe Medication Use Guidelines for the ICU.
5. Provide safety measures for drug interaction detection and prevention.
6. Develop a drug formulary proposal.

Abbreviations in This Chapter

ADE	Adverse drug event
ADR	Adverse drug reaction
CDS	Clinical decision support
EHR	Electronic health record
P&T	Pharmacy and therapeutics (committee)
QALY	Quality-adjusted life-year

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

Questions 1 and 2 pertain to the following case.
Your institution wants to evaluate the addition of coagulation factor Xa (recombinant), inactivated-zhzo to your formulary for reversal of direct oral anticoagulants.

1. Which best describes the type of cost this medication would have?
   A. Fixed medical.
   B. Variable medical.
   C. Indirect.
   D. Incremental.

2. Your institution wants to use pharmacoeconomic data to evaluate the impact of this medication on neurologic disability and cost compared with prothrombin complex concentrate. Which analysis would be best to find in the literature?
   A. Cost-benefit.
   B. Cost-effectiveness.
   C. Cost-minimization.
   D. Cost-utility.

3. A 60-year-old patient is admitted to the surgical ICU with a documented history of cefuroxime allergy (shortness of breath). The patient is administered ceftriaxone and develops a life-threatening anaphylactic reaction. Which best describes this patient's reaction to ceftriaxone?
   A. Adverse drug reaction (ADR).
   B. Side effect.
   C. Adverse drug event (ADE).
   D. Preventable ADE.

4. A 50-year-old patient is admitted to the neurology ICU from the medical floor for the management of an apixaban-induced cerebral hemorrhage. Which medication would best have been used as an effective tracer medication to help detect this ADE?
   A. Coagulation factor VIIa.
   B. Phytonadione.
   C. Protamine.
   D. Prothrombin complex concentrate.

Questions 5 and 6 pertain to the following case.
As a new pharmacist in the medical ICU at your institution, you have been tasked with improving the safety of medication use in the ICU. You have read the 2017 SCCM Guidelines for Safe Medication Use in the ICU and are prepared to provide recommendations to the interdisciplinary critical care committee regarding your suggestions.

5. According to the grading system of the guideline recommendations, which is the best initial recommendation to improve medication use safety, given the strength of recommendation?
   A. Fully integrating the institution's electronic health record (EHR) systems to improve communication.

B. Integrating interruptive clinical decision support (CDS) into the EHR.
C. Creating a policy requiring that intravenous medications can only be prepared by pharmacists.
D. Investing in smart intravenous infusion pumps to reduce medication errors.

6. After your institution has integrated your recommendation into routine medical care, a former patient recommends including the patient and other caregivers in reducing medication errors. According to the available literature, which best supports this recommendation?
   A. The 2017 SCCM guidelines provide a strong recommendation for this integration into routine care.
   B. Areas in the hospital outside the ICU provide data supporting this integration.
   C. Data analyses on direct observations of interactions between the medical team and patients have shown a reduction in medical errors in the ICU.
   D. Data analyses on chart reviews of ICU patients have shown that these reported outcomes are not routinely accounted for.

7. A 50-year-old patient was initiated on meropenem for empiric coverage of pneumonia, given prior culture data. They were subsequently initiated on valproic acid for the management of agitation; however, this was not effective for management and resulted in self-extubation and subsequent reintubation. In investigating this adverse event, which method would best evaluate the likelihood of its occurrence?
   A. Drug interaction database.
   B. Drug interaction probability scale.
   C. Naranjo nomogram.
   D. Roussel Uclaf Causality Assessment Method (RUCAM).

8. Your hospital wants to add meropenem/vaborbactam, a broad-spectrum antimicrobial targeting highly drug-resistant organisms that is FDA approved for complicated UTIs, to the formulary. Which would be the most appropriate restriction to its use?
   A. Restriction to intensivist use.
   B. Restriction to patients with hospital-acquired pneumonia.
   C. Restriction to infectious diseases clinicians' approval.
   D. Restriction to patients with appropriate cardiac monitoring.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 2: Therapeutics and Patient Management
   a. Task A: 1, 5–7
   b. Task C: 2, 4, 5
2. Domain 3: Professional Practice
   a. Task A: 1-6
   b. Task B: 1

I. PHARMACOECONOMICS

A. Overview
   1. Pharmacoeconomics is the description and analysis of the costs of drugs and pharmaceutical services as well as their effects on individuals, health care systems, and/or society.
   2. Economic evaluations identify, measure, and compare the costs and consequences of a pharmaceutical product or service. These can convey different results, depending on disease state/therapy being evaluated, alternative therapies, perspective (e.g., payer, provider, patient, society), and intended outcome.
      a. Full pharmacoeconomic analyses include the following:
         i. Cost-benefit
         ii. Cost-effectiveness
         iii. Cost-minimization
         iv. Cost-utility
      b. Humanistic outcomes or patient-reported outcome measures include the following:
         i. Functional status
         ii. Health status
         iii. Health-related quality of life
         iv. Patient satisfaction
         v. Quality of life

B. Types of Costs
   1. Costs versus charges
      a. Although the term *charges* alone is often used in the literature, *charges* are actually costs (also termed *expenses*) plus profit or a combination of the acquisition cost for a specific item used (e.g., cost of medication dose administered) plus *estimated* costs to provide the service (e.g., workflow and personnel associated with the process for medication use) plus a *markup.*
   2. Types of cost (Table 1)

Table 1. Summary of Types of Cost Used in Economic Evaluations

Type of Cost	Definition	Examples
Direct • Medical	Costs associated with prevention, detection, or treatment of a disease or illness Costs associated with medical care	**Fixed** (costs that remain constant and are not dependent on volume of use; not typically included in pharmacoeconomic evaluations): Rent, salaried workers **Variable** (costs depend on volume of use): Laboratory tests, medications, procedures
• Non-medical	Costs as a result of disease or illness that do not involve the purchase of medical services	Child care, specialty diets, transportation
Indirect	Costs associated with morbidity or mortality, related to change in productivity as a result of disease or illness	Costs because of inability to work, income lost because of premature death
Intangible	Costs associated with nonfinancial outcomes of disease	Costs from posttraumatic stress disorder from ICU survival; post-ICU syndrome
Incremental	Additional costs to purchase additional benefit or effect of medical care	Additional medications to control atrial fibrillation that started in ICU
Total	Inclusion of all the above costs	N/A

N/A = not applicable.

C. Types of Pharmacoeconomic Evaluations (Table 2)
   1. There are several types of full economic evaluation. The main difference between the types of full economic evaluation is how the benefits to the individual are measured and valued.
   2. Cost analysis
      a. Compares only cost (i.e., no effectiveness) of two or more treatments or services
      b. Potentially more applicable than cost-minimization analysis when effectiveness is not known
   3. Cost-benefit analysis
      a. Systematic approach assessing strengths and weaknesses of treatments or programs to determine the best approach to attaining benefits while minimizing expense
      b. Both costs and outcomes are measured in monetary units, including direct and indirect benefits, allowing for a direct comparison.
      c. Identifies yield of investment (e.g., justification of post-ICU clinic, comparison of a pharmacokinetics service with a vaccination program)
      d. Results may be expressed with cost-benefit ratio, net cost, or net benefit.
   4. Cost-effectiveness analysis
      a. Compares two or more treatment alternatives or programs in which resources (e.g., treatments, services) are used
      b. Costs are measured in monetary terms, whereas outcomes are measured in units of effectiveness.
      c. Determines which alternative can produce the desired effect or maximization of effect, which may by a more expensive option that provides a more desirable outcome than a less expensive option
      d. Results are expressed as the incremental cost-effectiveness ratio (ICER), which is defined as the additional cost to the next most effective intervention producing another unit of output. These results can be displayed in a cost-effectiveness plane, which graphs the ICER in four quadrants (x-axis is effectiveness; y-axis is cost).
   5. Cost-minimization analysis
      a. Compares two or more treatment alternatives or programs for which the outcomes are virtually identical (e.g., effectiveness) except for costs; therefore, results are expressed in monetary form, which allows for ease in examining resource outcomes
      b. Costs are compared to determine the least expensive alternative with a noninferior efficacy.
      c. Should not be used if no evidence supports the effectiveness of the treatment alternative
   6. Cost-utility analysis
      a. Comparison of two or more treatment alternatives or programs in which costs are measured in monetary terms and outcome is expressed with respect to patient preferences, quality of life, or quality-adjusted life-years (QALYs)
         i. Form of cost-effectiveness analysis in which values (utilities) are assigned to the outcome
         ii. Life and death are reference states for the utilities, with perfect health = 1.0 and death = 0.0 (although to a patient, a certain state [e.g., vegetative] may be worse than death, allowing for a negative utility).
         iii. Utilities or a patient's preference for a disease state can be measured by obtaining information from the literature, approaching a convenience sample of experts, using decision theory with a direct approach, or using psychometric methods with indirect approaches.
            (a) Direct approaches include rating scales, time trade-off (acute and chronic conditions), and standard gamble (acute and chronic conditions).
            (b) Indirect approaches include multi-attribute health state scores (e.g., quality-of-life assessment tools).
      b. To calculate QALYs, the life-years gained is determined and then multiplied by the utility of that life-year. For example, if persons have a life expectancy of 20 years but have a disability and believe they are functioning at only 70% of their capacity, the QALY is 20 years × 0.7 = 14 QALYs.
      c. Measures the function and overall well-being of patients using quality-of-life assessment tools, such as those used in the ICU, which are general and/or disease-specific (e.g., EuroQol 5D Questionnaire, European Organization for Research and Treatment of Cancer 8 Dimensions)

Table 2. Summary of Pharmacoeconomic Evaluations[a]

Economic Evaluation	Cost Measurement	Consequences or Outcomes	Measurement/Valuation of Consequence or Outcomes	Example	Challenges
Cost analysis (partial evaluation)	Monetary	Not considered, so not a full economic evaluation	Not considered, so not a full economic evaluation	Cost of cisatracurium compared with rocuronium in patients with acute respiratory distress syndrome (disregarding any outcome)	Challenges limited compared with CMA, CBA, CEA, CUA
Cost-benefit analysis (CBA)	Monetary for total cost of care	Single or multiple effect/outcome that does not have to be common to both alternatives Effects are translated to monetary units, so comparison among various alternatives is possible	Monetary units	Comparison of pharmacist-led blood factor prescription with standard of care on andexanet costs to hospital	Translating consequences to monetary units
Cost-effectiveness analysis (CEA)	Monetary for total cost of care	Must have a single effect/outcome of interest for all alternatives considered, and the effect/outcome is achieved to a different degree The outcomes of the alternative strategies are not equivalent and are measured in unidirectional natural units	Nonmonetary physical units of effectiveness. (e.g., life-years gained, surrogate end point)	Comparison of andexanet with 4F-PCC on bleeding-related mortality	Determining the probabilities of the clinical pathway for the decision tree to determine the consequences
Cost-minimization analysis (CMA)	Monetary for total cost of care	Must be evaluated Single effect/outcome of interest for all alternatives that were shown to be equivalent	Identical	Comparison of medication and its biosimilar on cost because equivalent outcomes are assumed	Proving consequences are identical
Cost-utility analysis (CUA)	Monetary for total cost of care	Single or multiple effect/outcome that does not have to be common to both alternatives Effects are translated to QALYs, so comparison among various alternatives is possible	Utility values and QALYs	Comparison of andexanet with 4F-PCC on patient outcomes, measured by patient utility (QALY)	Measuring utilities

[a]Andexanet = coagulation factor Xa (recombinant), inactivated-zhzo.
4F-PCC = 4-factor prothrombin complex concentrate; QALY = quality-adjusted life-year.
Modified from: Drummond MF, Sculpher MJ, Claxton K, et al. Methods for the Economic Evaluation of Health Care Programmes, 4th ed. Oxford University Press, 2015.

D. Application of Pharmacoeconomics
   1. Formulary management to determine whether newly marketed or approved medications should be included/excluded with/without use criteria that may affect prescribing patterns
   2. Development of clinical guidelines, policies, or protocols to promote the most cost-effective and desirable use of medications in the context of safety and efficacy
   3. Implementation of drug use policy to promote the most efficient use of health care products and services
   4. Perform program evaluations to determine the value of an existing medical or pharmacy service or the potential worth of starting a new service, in addition to guiding the establishment of key business objectives or metrics of growth, adoption, or process control
   5. Population health analytics (e.g., use of pharmacoeconomic analyses to determine evidence-based medicine treatment decisions within a health system)
   6. Determine individual patient treatment decisions
   7. Publication of pharmacoeconomic evaluations with pharmacist involvement

Patient Case

1. A 30-year-old patient is admitted to the surgical ICU after a motor vehicle accident. Which would best exemplify an indirect cost from their perspective?
   A. Loss of wages from occupation as an engineer.
   B. Effects of posttraumatic stress disorder from the accident.
   C. Fentanyl administered to manage pain.
   D. Cost of the ambulance that transported them to the hospital.

II. DRUG-RELATED EVENTS: MEDICATION ERRORS, ADES AND ADRS, PREVENTABLE ADES

A. Overview
   1. In the United States, 770,000 injuries or deaths are caused by ADRs annually; the cost burden of ADEs is greater than $5.6 million per hospital.
      a. Up to 60% of hospital readmissions are attributable to either medication access issues or ADEs – This was before the data explosion with the EHR, which may have made it worse if it is used in discordant fashion.
      b. A severe ADE increases the hospital length of stay by 8–12 days at a cost of $16,000–$24,000 per patient.
      c. The cost burden of preventable ADEs in the United States is $4.1 million per hospital annually. Medical malpractice lawsuits are often related to, or occur because of, ADEs.
   2. The risk of ADEs increases when patients are critically ill, have a higher chance of being administered a high-risk medication, and have altered response to medications because of altered pharmacokinetics and organ failure.
   3. The Joint Commission medication management standard requires hospitals to respond to actual or potential ADEs, significant ADRs, and medication errors. At the very least, hospitals need to respond, document and report, and manage ADEs. The pharmacist, who is the drug expert, should play an integral role in prospective order verification to prevent ADEs by intervening as needed.

B. Definitions of Drug-Related Events
   1. Figure 1 summarizes drug-related events.

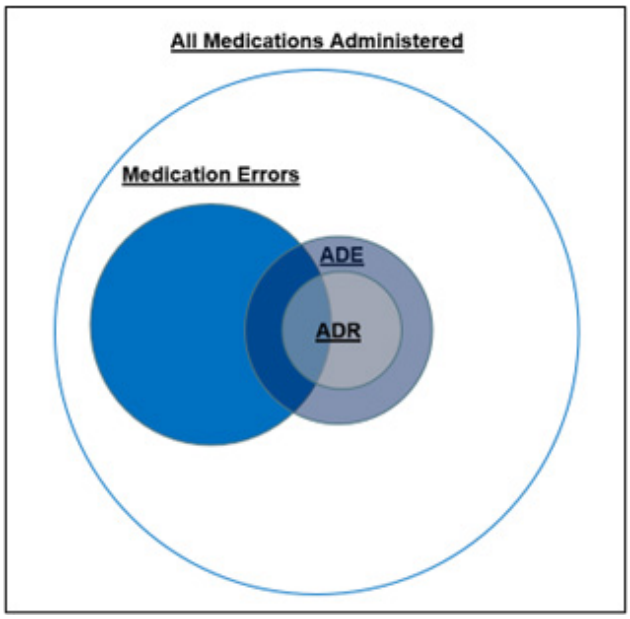


Figure 1. Types of drug-related events.

This figure details types of drug-related events including medication errors, adverse drug events (ADEs), and adverse drug reactions (ADRs). The largest circle illustrates all medications administered, with the next largest circle indicating which medications administered were associated with a medication error that reached the patient. The third largest circle illustrates ADEs, which may be a result of medication errors (i.e., overlap between circles). ADEs that are a result of medication errors are therefore preventable. The smallest circle illustrates ADRs, which are a type of ADE that causes patient harm/injury. Some of these are the result of medication errors, as suggested by the overlap between the "Medication Errors" and "ADR" circles.

   2. Medication errors
      a. Mishaps that occur during the medication use process (e.g., reconciling, prescribing/ordering, transcribing, dispensing, administering, monitoring)
         i. Medication errors that are intercepted and stopped before they occur are called "near misses."
         ii. Some medication errors cause harm (result in an ADE), and some do not. Medication errors should be reported through the health system's patient incident reporting system. Harm is physical or psychologic injury or damage to a patient's health, including both temporary and permanent injury.
      b. Medications errors can be categorized as the following:
         i. Category A: No error – Scenarios that can cause error
         ii. Category B: An error occurred but did not reach the patient
         iii. Category C: An error occurred and reached the patient but did not cause harm
         iv. Category D: An error occurred that reached the patient and required monitoring to confirm that it resulted in no harm to the patient or required intervention to prevent harm
         v. Category E: An error occurred that may have contributed to or resulted in temporary harm to the patient and required intervention
         vi. Category F: An error occurred that may have contributed to or resulted in temporary harm to the patient and required initial or prolonged hospitalization
         vii. Category G: An error occurred that may have contributed to or resulted in permanent patient harm

viii. Category H: An error occurred that required intervention necessary to sustain life
ix. Category I: An error occurred that may have contributed to or resulted in the patient's death

c. Medication errors are defined by The Joint Commission as a sentinel event that includes events such as those that cause severe temporary harm (requiring intervention), permanent harm, or death (e.g., potentially category E, categories F–I).
d. May be reported through the Institute for Safe Medication Practices (ISMP Medication Error Reporting Program)
   i. A confidential and voluntary system, investigated by ISMP staff and then reported to the FDA's MedWatch program and product vendors
   ii. A separate reporting method is used for vaccines, specifically the ISMP Vaccine Errors Reporting Program. Vaccine medication error reports provide an option to submit the vaccine's manufacturer name, dosage, lot number, expiration date, and National Drug Code.
   iii. Consumers may also report medication errors through ISMP.
   iv. Medication errors may be reported directly to the FDA by the MedWatch system. The FDA's MedWatch allows ADE and medication error reporting.

3. Adverse drug events
   a. *Injuries* resulting from use of a drug, which include harm caused by the drug (ADRs and overdoses) and harm from use of the drug, including dose changes and discontinuations of drug therapy
   b. Overall, *ADE* is the broader term used to describe any harmful event associated with a medication, including inappropriate use such as an overdose, whereas *ADR* is used when an adverse response, including harm, occurs with normal use of the medication.
4. Preventable ADEs
   a. A preventable ADE occurs when a breach of standard professional behavior, practice, or technique was identified or when necessary precautions were not taken, or when the event was preventable by modifying behavior, practice, technique, or care.
      i. Results from any medication error that reaches the patient and causes harm
      ii. About 30%–50% of all ADEs are preventable.
      iii. Drug interactions account for 3%–5% of all preventable in-hospital ADRs (drug interactions are discussed further in section IV, Drug Interaction Surveillance and Prevention).
   b. Examples of preventable ADEs
      i. Carbamazepine prescribed for an Asian patient with bipolar disorder without testing for the *HLA-B*1502* allele; in turn, the patient develops carbamazepine-induced toxic epidermal necrolysis
      ii. Heparin administered without the use of weight-based dosing, causing a supratherapeutic partial thromboplastin time and an intracranial hemorrhage
      iii. Phenytoin mixed accidentally with dextrose rather than normal saline, causing precipitation and lack of drug potency and leading to a patient developing withdrawal seizures and status epilepticus
5. Nonpreventable ADEs
   a. ADEs not associated with a medication error and are unintended reactions without known mitigation strategies resulting in patient harm (a patient with a bleed despite appropriate dosing, administration, and monitoring of the anticoagulant)
6. Adverse drug reactions
   a. Definitions
      i. WHO defines ADRs as "any response to a drug which is noxious and unintended, and which occurs at doses normally used for prophylaxis, diagnosis, or therapy of disease, or for the modification of physiological function."

ii. According to Karch and Lasagna (JAMA 1975;234:1236-41), ADRs are "any response to a drug that is noxious and unintended, and that occurs at doses used in humans for prophylaxis, diagnosis, or therapy, excluding failure to accomplish the intended purpose."
iii. The primary difference between the WHO and Karch and Lasagna ADR definitions is that, according to the WHO definition, therapeutic failures are an unintended effect and an ADR, whereas according to the definition by Karch and Lasagna, they are not. Both definitions refer to doses normally used in humans, which exclude overdoses and medication errors that exceed normal doses.
iv. ADRs caused by therapeutic failures with easily retrievable, detailed data that document the drug and the therapeutic failure can significantly affect patient care.

b. Examples by definition
i. WHO
(a) Clopidogrel failure to prevent an ischemic stroke
(b) Enoxaparin failure to prevent deep vein thrombosis
(c) Famotidine failure to prevent intravenous ketorolac-induced gastropathy
(d) Pantoprazole failure to prevent GI bleeding from a stress ulcer in a critically ill patient
ii. Karch and Lasagna
(a) Lorazepam when used to treat anxiety may cause sedation.
(1) This is an unintended adverse reaction; conversely, lorazepam for sedation in a critically ill patient is an intended effect, not an ADR.
(b) Diphenhydramine causing unintended sedation when used to treat allergic rhinitis
(1) This is an ADR; conversely, diphenhydramine as a sedative hypnotic is an intended effect, not an ADR.

C. FDA-Reportable ADEs
1. For reporting an ADE to the FDA, the FDA defines ADEs as serious adverse events related to drugs or devices in which "the patient outcome is death, life threatening (real risk of dying), hospitalization (initial or prolonged), disability (significant, persistent, or permanent), congenital anomaly, or required intervention to prevent permanent impairment or damage." FDA-reportable serious ADEs may include the following:
a. Acetaminophen-induced hepatotoxicity
b. Apixaban-induced intracranial hemorrhage
c. Gentamicin-induced irreversible ototoxicity
2. The FDA is also interested in serious ADE reports for newly marketed drugs or novel adverse events that have not previously been reported for new or old drugs.
a. The FDA Adverse Reporting System (FAERS) database is designed as a postmarking surveillance tool for medications and biologics. Other therapies, such as medical devices, have their own reporting system.
b. The CDC and FDA co-sponsor the Vaccine Adverse Event Reporting System (VAERS), which is focused on identifying potential signals for adverse events associated with vaccines licensed for use within the United States.
3. The FAERS and VAERS databases are used as postmarketing surveillance systems. Limitations of FAERS and VAERS include the following:
a. Lack of certainty of adverse event (i.e., may not be validated)
b. Underreporting
c. Less severe adverse events likely to be underreported
d. Potential lack of detail provided

4. Reporting of ADEs by health care professionals and consumers to the FDA is voluntary and may be done through the FDA's MedWatch program (begun in 1993). All of the following may be reported with respect to an FDA-regulated medication, biologic, tobacco product, dietary supplement, cosmetic, or medical device:
   a. Serious ADE
   b. Serious ADR
   c. Product quality problem
   d. Product use error
   e. Therapeutic inequivalence
   f. Therapeutic failure
5. Clinicians may report an ADE directly to the drug's manufacturer. The pharmaceutical manufacturer has a legal responsibility to follow up with the reporter on all ADEs reported and to report the ADE to the FDA.
6. ADE reports and medication error reports submitted directly by health care professionals or manufacturers are entered in the FAERS database.
7. If the FDA detects a safety concern, regulatory action may be needed to protect the public, such as updating the product labeling, restricting the drug, communicating the safety concerns to health care professionals and consumers, or removing the drug from the market.

D. Designing an ADE Reporting Program
   1. A comprehensive ADE program should have a policy and procedure, with guidelines for ADE detection, reporting, management, surveillance, and education. Appendix 1 shows an ADE reporting form.
   2. Because of pharmacists' expertise in drug-induced diseases and their role in preventing and managing ADEs, pharmacists often serve as chair or co-chair of the ADE committee.
   3. The ADE committee should be interdisciplinary and composed of the following:
      a. Prescribers
      b. Pharmacists
      c. Nurses
      d. Risk management personnel
      e. Quality assurance and performance improvement personnel
      f. Other health care providers
   4. In general, the ADE committee is a subcommittee of the pharmacy and therapeutics (P&T) committee that reports to the P&T committee.
   5. The ADE committee should meet monthly, bimonthly, or quarterly, depending on the number of reports and actionable items that need review.
   6. ADE data can be reported by a specialty unit or service (e.g., emergency medicine, ICU).
   7. ADE data can ensure that appropriate preventive measures are developed for that specialty unit.
      a. A proactive method to reduce the incidence of ADEs is a failure mode and effects analysis (FMEA), which is a proactive, structured method to identify potential failure points within a system that may lead to an ADE.
      b. If an ADE has occurred, performance of a root cause analysis, which is a structured method to determine why an ADE occurred (i.e., retroactive), can help identify methods to prevent the recurrence of the ADE in the future.
   8. ADR and ADE data should be reported as mild, moderate, or severe. Many scales are available in the literature. Definitions for each should be established. Box 1 provides an example.

Box 1. Definitions for the Degree of ADR or ADE Severity[a]

1. Mild ADR or ADE: Results in heightened need for patient monitoring with or without a change in vital signs, but no ultimate patient harm, or any adverse event that results in the need for increased laboratory monitoring
2. Moderate ADR or ADE: Results in the need for aggressive intervention with antidotes or increased length of hospital stay (e.g., severe hypotension [e.g., BP < 90/50 mm Hg], bleeding necessitating transfusions)
3. Severe ADR or ADE: Results in harm to the patient, prolonged hospitalization, transfer to a higher level of care, permanent organ damage (e.g., irreversible hepatotoxicity or renal failure), or death with probable ADE causality nomogram score

[a]Other ADE severity scales are available.
ADE = adverse drug event; ADR = adverse drug reaction; BP = blood pressure.

9. The ADE committee should
    a. Designate which ADEs are preventable, explain why they are preventable, and determine possible preventive measures
    b. Determine which ADEs will be reported to the FDA or the manufacturer
    c. Report the data regarding who is detecting ADEs and who reports, documents, and manages the ADEs
    d. Provide trending data on the basis of either drug or drug class and by specialty units
    e. Benchmark the hospital's ADEs against itself in previous years and compare them with the data from other hospitals published in the biomedical literature. Total ADE data can be reported according to the following:
        i. Total number of ADEs
        ii. ADEs per admission
        iii. ADEs per patient
        iv. ADEs per patient-days
        v. ADEs per doses dispensed
        vi. ADEs per doses administered
10. A popular method of reporting ADEs is by the total number of admissions, with an acceptable benchmark of 2.5%–10% (e.g., a hospital with 10,000 admissions reporting 1000 ADEs would have a 10% ADE reporting rate). ADE benchmarks are difficult to determine because of the many variables that affect the reporting methods, such as the ADE definition used by the facility or the definition used by the reporter, the number of clinical pharmacists available to report, the vigilance and emphasis of the reporting systems, and the use of technology or reports to increase reporting.

E. Documenting ADRs and ADEs in the Patient's Medical Record
    1. Allergy data are always documented in the patient's medical record and are required in the patient profile. Preferably, the specific medication that caused the allergy, the type of reaction, and when the allergy occurred should be documented.
    2. The EHRs where these data are often found were incentivized by "meaningful use." The Centers for Medicare & Medicaid Services program is meant to assist in adopting EHR technology, including advanced data sharing and a focus on quality of care. In 2018, the term *meaningful* use was changed to *promoting interoperability.*

3. A drug-induced allergy can be an ADR or an ADE, depending on severity.
    a. It is paramount to report and document ADE data in the patient's medical record for the same reason as documenting allergy data to prevent recurrence with the same drug or a drug from the same or similar drug class, and to mitigate risk when the same drug may need to be used again or to mitigate risk when other medications that can cause the same adverse event are used.
    b. If possible, a true allergy and an ADE/ADR should be differentiated between within the EHR.
4. ADR and ADE data should be recorded in the electronic medical record and maintained in the record indefinitely.
    a. For mild ADRs: When the same medication is prescribed, the CDS system can be set to alert the prescriber, or the data can be retrievable for review but without prompting the clinician (e.g., lisinopril-induced hypotension).
    b. For moderate ADRs: When the same drug is prescribed, the EHR, including the CDS system, can be programmed to highlight or warn the prescriber or require a text response with an explanation for its continued use.
    c. For severe ADRs and ADEs: When the same drug is prescribed, the EHR, including the CDS system, can be programmed to cause a hard stop and prevent the drug from being prescribed or a hard stop requiring a prompt with an explanation for use (e.g., isoniazid-induced hepatotoxicity).

F. Methods of Medication Error and ADR or ADE Surveillance
1. ADEs can be detected prospectively and retrospectively.
    a. Pharmacists may detect ADEs prospectively while on patient care daily rounds, or from communication with patients while administering medication histories or transitions of care (e.g., ICU to step-down) or during discharge counseling.
    b. An example of CDS is automated triggers used for prospective surveillance of events (i.e., before an event has occurred). Pharmacists may also detect ADEs retrospectively through medical record reviews or during medication histories or chart reviews (i.e., after an event has occurred, so limited opportunity for intervention).
2. Although CDS is reinforced by the literature on reducing medication errors, it must be weighed against the potential for alert-driven interventions to cause "alert fatigue."
    a. No consensus definition exists for alert fatigue; however, it can be conceptualized as how increasing the number of alerts may result in decreased attention to interacting with the alerts.
    b. Given the lack of definition, it is difficult to measure alert fatigue. Potential metrics may include the following:
        i. Sensitivity, specificity, positive predictive value, and negative predictive value of alerts
        ii. Number of alerts overridden
        iii. Number of alerts inappropriately overridden
        iv. Patient harm associated with inappropriately overridden alerts
    c. This fatigue is associated with an increased risk of preventable ADEs in a critically ill population.
    d. Further research is needed on how to use alerts and other types of CDS more effectively, specifically in the ICU.
3. Retrospective incident voluntary reporting is an ethical obligation of all health care professionals.
    a. Voluntary reporting is the primary source of event identification for most institutions; however, events are grossly underreported.
    b. Typically, safety pharmacists are responsible for aggregating these data. These aggregate reports are reviewed by the ADE committee for the institution. Possible prevention methods are determined by the committee. This is a retrospective evaluation because events are typically evaluated when there is allotted time, which is often when the patient is discharged from the hospital. This can be a prospective method if real-time evaluation is an option.

c. Trigger/tracer drugs
   i. ADEs can be detected using a CDS system within an EHR for *trigger* or *tracer drugs* – these terms are used synonymously. Trigger or tracer drugs are drugs that are routinely prescribed to treat ADEs, such as antidotes, or physiologic antagonists or agents for gastric decontamination. These triggers prompt a targeted medical record review. Box 2 provides examples.
   ii. The CDS system can be programmed with a list of tracer drugs so that when prescribers order a tracer drug, they are asked whether the order is for an ADE and, if so, what type of ADE occurred. The potential ADE can then be reported and managed by the clinical pharmacist.
   iii. One of the benefits of this system is that it captures ADEs that might not have been reported, especially when a clinical pharmacist may not be present on the unit or on daily patient care rounds, and it allows for more physician-reported ADEs.

Box 2. Examples of Medications Used as Triggers or Tracers to Aid in Reporting Adverse Drug Events

1. Acetylcysteine (enteral, injection)
2. Activated charcoal suspension
3. Atropine (injection)
4. Benztropine
5. Dantrolene
6. Diphenhydramine
7. Diphenoxylate/atropine
8. D50W
9. Digoxin immune FAB
10. Epinephrine 0.15/0.3 mg autoinjector
11. Fidaxomicin
12. Flumazenil
13. Glucagon
14. Hydrocortisone (injection, topical)
15. Loperamide
16. Naloxone
17. Phytonadione
18. Prednisone
19. Protamine
20. Prothrombin complex concentrate
21. Sodium zirconium cyclosilicate
22. Vancomycin (enteral, rectal)

$D_{50}W$ = dextrose 50% in water.

d. Therapeutic drug concentration and abnormal laboratory value monitoring can be another source for ADE detection. The pharmacy may design daily reports that contain all the abnormal drug serum concentrations such as the phenytoin, digoxin, lithium, vancomycin, and aminoglycoside concentrations, or other laboratory monitoring values (e.g., lisinopril and potassium). The clinical pharmacist can then review all cases of supratherapeutic concentrations and laboratory abnormalities by performing a targeted medical record review for ADEs. Although these are basic CDS functions that include medications and laboratory values as the knowledge for the alert, more advanced, "smarter" alerts are being used that include risk scoring–based approaches that help prioritize the pharmacist's reviews.

e. Identified events must be confirmed for causality. Many instruments can help link the drug with the event. Structured instruments create a more reliable and valid assessment. A sample ADE form with the Naranjo criteria for causality determination is provided in Appendix 1. This is the most commonly used ADR causality instrument, though its reliability and validity in the critically ill population need improvement. Another instrument specific for drug-induced liver injury is the RUCAM. Causality assessment includes temporal sequence, dechallenge (removal of the suspect drug), rechallenge (reintroduction of the suspect drug), evaluation of other causes, objective evidence that is available or obtained, and history of a reaction to a similar drug.

4. Prospective
   a. Direct observation
      i. One method of medication error detection is through direct observation; this can be accomplished using the medication pass method, in which the pharmacist observes nurses in the medication administration process and notes any errors that occur.
      ii. Direct observation provides the unique advantage of capturing medication administration errors that are not typically identified with other detection methods. Other examples of direct observation include nurses observing nurses or physicians observing physicians during medication administration. This is considered prospective because observation occurs in real time but errors are evaluated later.
      iii. A potential limitation of direct observation is that if an individual is aware of being observed, the individual may change behavior to be "optimal" (i.e., the "Hawthorne effect").
   b. CDS with alert generation can be used for prospective or real-time surveillance.
      i. Prescribing – Providers can be notified of high-risk scenarios during the ordering process in order to prevent medication errors, such as prescribing enoxaparin for a patient with a CrCl less than 30 mL/minute. Alerts during medication ordering should be used judiciously and created with greater precision to avoid alert fatigue.
      ii. Order verification – Pharmacists receive preventive alerts during order verification to avoid medication errors and the potential for ADEs (e.g., drug-drug interactions, duplicate therapy).
      iii. High-risk scenarios in real time – Pharmacists may receive advanced alerts outside order verification, indicating when a patient is initiated on a drug, but the patient's clinical scenario presents a risk (e.g., patient receiving an epidural and initiated on a systemic anticoagulant).

Patient Cases

2. Which best classifies the degree of severity of an ADE in a patient who develops enalapril-induced asymptomatic hyperkalemia with no noted ECG changes (K 5.7 mEq/L) managed with one dose of sodium zirconium cyclosilicate?

   A. Mild.
   B. Moderate.
   C. Severe.
   D. Life threatening.

3. Which best classifies the degree of severity of an ADE in a patient who is in the geriatric psychiatry unit and develops intravenous haloperidol-induced torsades de pointes that is successfully treated with intravenous magnesium and managed with additional monitoring in the cardiac care unit and telemetry?

   A. Mild.
   B. Moderate.
   C. Severe.
   D. No classification.

Patient Cases *(continued)*
Questions 4 and 5 pertain to the following case.
M.S., a 77-year-old patient residing in a nursing home, has taken lisinopril 10 mg daily for the past 3 months. They are admitted to the ICU with lisinopril-induced angioedema and present with severe tongue swelling, stridor, and shortness of breath that necessitated a tracheotomy. They had no history of allergies and did not miss any doses of lisinopril.

4. Which best describes this patient's reaction to lisinopril?
   A. ADE.
   B. Preventable ADE.
   C. Medication error.
   D. Preventable ADR.

5. Which best classifies the degree of severity of this patient's ADE to lisinopril?
   A. Mild.
   B. Moderate.
   C. Severe.
   D. No classification.

III. 2017 SCCM CLINICAL PRACTICE GUIDELINE: SAFE MEDICATION USE IN THE ICU

A. Structure of Guideline
   1. Developed in 2017 with literature searches up to the end of 2015, focused on 34 PICO (population/problem/patient, intervention/indicator, control/comparator/comparison, and outcomes) questions, including data on adult and pediatric ICU patients
   2. First clinical guideline to address safe medication use in the ICU. Identified that rigorous research to guide practice is still needed in many areas
   3. Three key components: environment and patient, medication use process, and patient safety surveillance systems
   4. Recommendations based on GRADE system (quality of evidence: A = high to D = very low; strength of recommendation: 0 = no recommendation, 1 = strong, 2 = weak)

B. Highlights of Environment and Patient
   1. ICU patients have different risk factors for ADEs, and the severity of medication errors is higher than with general care (non-ICU) patients (both grade C).
   2. Guidelines suggest that educational efforts should be used to reduce medication errors in the ICU (2C).
   3. No recommendation was provided regarding disclosure of medication errors/ADEs to patients and/or family members, because of lack of data, but this is recommended by the National Quality Forum.

C. Highlights of Medication Use Process
   1. Four stages of process (prescribing, dispensing, administration, monitoring)
   2. Prescribing
      a. Guidelines suggest implementing computerized provider order entry to decrease medication errors and preventable ADEs (2B). Some data support medication error reduction, but evidence is conflicting, with a decrease in omission errors but an increase in duplicate order errors.

b. Guidelines suggest using CDS systems to decrease the number of medication errors and ADEs (2C).
   i. CDS systems are tools designed to enhance clinician decision-making (e.g., allergy, drug-drug interaction, dosing recommendations).
   ii. Trigger-initiated CDS alerts cause a concern for alert fatigue that requires a balance between sensitivity and specificity because of lack of patient-specific considerations (e.g., ICU level of care).
c. Guidelines suggest using protocols/bundles to reduce medication errors/ADEs (2B).
d. No recommendation is provided regarding the use of medication reconciliation to decrease medication errors and ADEs in ICU patients or for transitions of care.

3. Dispensing
   a. Guidelines recommend compliance with safe medication concentration practices (e.g., premade mixes, standardized medication concentrations within institution) to reduce the number of medication errors and potential ADEs (1B).
   b. Guidelines suggest use of automated packaging and dispensing of medications to reduce medication errors/ADEs versus non-automated methods (both 2C).
   c. Guidelines suggest use of medication labeling practices (e.g., tall man lettering) for sound-alike look-alike drugs to reduce medication errors (2B).
4. Administration
   a. Guidelines suggest use of bar code medication administration (BCMA) to reduce medication errors and/or ADEs in the ICU (2C).
      i. Conflicting evidence on benefit, though data largely suggest reduction in medication errors and potential ADEs
      ii. One limitation of BCMA is that workarounds will reduce effectiveness.
   b. Guidelines suggest smart intravenous infusion pumps be used to reduce the rate of medication errors and ADEs (2C).
      i. Evidence is conflicting on the use of the guardrails and the types of errors evaluated.
      ii. Concern for auditory alarm fatigue exists with smart pumps, especially in the context of other auditory alarms in the ICU.

D. Highlights of Patient Safety Surveillance Systems
1. Guidelines recommend use of direct observation as an active medication surveillance system to identify the medication errors (1A).
2. Guidelines suggest nontargeted chart review to detect ADEs, versus voluntary reporting, to increase the number of ADEs identified, though this is a resource-intensive process (2C).
3. Guidelines suggest targeted chart review to increase ability to detect ADEs compared with voluntary reporting (2B). Strategies include evaluation of only a specific section of the patient's medical record (e.g., ICU discharge note) or trigger medications.
4. Guidelines suggest patient and family reporting at or after ICU discharge to improve medication error and ADE detection (2C). This includes formalization of a process for interviewing patients or family members about possible medication errors or ADEs that occurred while the patient was in the ICU.

IV. DRUG INTERACTION SURVEILLANCE AND PREVENTION

A. Definition
   1. When the effects of one drug can be changed by the presence of another substance (e.g., medication, dietary), which may be intentional, clinically insignificant, or harmful and potentially life threatening
   2. Drug-related toxicity may be preventable and may require therapeutic intervention (e.g., monitoring, discontinuation of medication).

B. A documented drug interaction with known outcomes can be considered a medication error, a preventable ADE, or an ADR, depending on clinical context.

C. Drug infusion compatibility interactions (e.g., Y-site) are more common in the ICU because of increased likelihood of prolonged infusions.

D. Drug interaction databases include Lexicomp, IBM Micromedex, Epocrates, Clinical Pharmacology, Hansten and Horn's *Drug Interaction Analysis and Management,* Stockley's *Drug Interactions,* and *PDR Drug Interactions*. Many EHRs include Medi-Span and First Databank.
   1. Most compilation databases have drug interaction software in which the pharmacist can provide a list of drugs and the database will provide the type of interaction and severity. Of note, agreement between databases for severity of interactions is estimated to be less than 50%.
   2. Databases use different methods to assess severity of drug interactions.
   3. Clinical judgment should always be applied when assessing drug interactions.

E. Safety Measures to Avoid/Manage Drug Interactions
   1. Pharmacist review and validation
      a. Pharmacists are highly trained in pharmacology and drug interactions and are expected to apply advanced analysis and clinical qualification/management of drug interactions.
      b. Limitations include that not all medications a patient uses are filled by one pharmacy as well as the lack of potential data that can be pulled from a patient's EHR that can inform the existence and clinical relevance of drug-drug interactions.
   2. Using EHRs with CDS alerts that detect drug interactions
      a. Examples include passive pop-up warnings (e.g., alerts), active alerts with soft stops (e.g., documented allergy of nausea), and active alerts with hard stops (e.g., documented allergy of anaphylaxis).
      b. May prompt prescriber or pharmacist during order verification with a high frequency of alerts, leading to desensitization (i.e., alert fatigue)
      c. Despite the potential benefits of CDS, within the ICU patient population overrides are common, with a rate of around 80%, depending on the alert type. Inappropriate overrides of CDS alerts within the ICU are associated with an increased risk of patient harm.
      d. Drug-drug interaction alert fatigue can be managed by reducing the alert number, using severity or tiering of the interaction as a criterion for alert selection.
   3. Education as a method to prevent drug interactions (e.g., lectures, patient care rounds)

F. How to Evaluate Drug Interaction Cases
   1. Assessing causation of a drug interaction includes a temporal relationship, consideration of the pharmacologic properties of the object and precipitant drug, patient factors and disease states, the possible contribution of other drugs, and, when possible, a positive dechallenge. The temporal sequence is the key element in these considerations to help determine the cause.
   2. ADE nomograms such as the Naranjo nomogram are designed to evaluate ADEs, not drug-drug interactions; therefore, they should not be used to evaluate drug-drug interaction cases.

3. The drug interaction probability scale may be used to determine drug-drug interaction causation, including the adverse outcomes in a specific patient (Appendix 2).
    a. The drug interaction probability scale is patterned after the Naranjo ADR probability scale. A series of 10 questions related to the drug interaction are assessed with "yes," "no," "unknown," or "not available" answers and then scored and tabulated. The total score determines the probability of the drug-drug interaction occurring in the patient as follows:
        i. Highly probable: More than 8
        ii. Probable: 5–8
        iii. Possible: 2–4
        iv. Doubtful: 2 or less
    b. When using the drug interaction probability scale, the evaluator must have comprehensive knowledge of the pharmacologic properties of both the object and the precipitator drug, especially their pharmacokinetic and pharmacodynamic properties and their mechanisms of drug action and mechanisms of drug interactions, which can be a limitation to its use.

V. FORMULARY PROPOSAL

A. The drug formulary should be comprehensive and include all the medications needed for patient care.

B. The Joint Commission medication management standard requires hospitals to develop and approve the criteria for selecting medications into the drug formulary.

C. At a minimum, the drug selection criteria should include the following:
    1. Indications for use (FDA label approved and off-label)
    2. Efficacy and effectiveness
    3. Drug interactions
    4. Adverse effects
    5. Sentinel event advisories
        a. May include REMS (Risk Evaluation and Mitigation Strategies) for certain medications (if one exists)
        b. This FDA medication safety program is for medications with known significant safety concerns (e.g., birth defects, hepatotoxicity) that require an understanding of its risks and necessary interventions that are preventive to safely use the medication.
    6. Cost acquisition and total cost of care: This is a good application of the budget impact analysis. Ideally, if cost-effectiveness and cost-utility analyses are also available, they should be considered; however, they are typically not available this early in the drug approval process.

D. The P&T committee is responsible for maintaining the drug formulary and ensuring that new medications that can improve patient care are reviewed for formulary inclusion.

E. Critical care pharmacists should be cognizant of new medications and new medication dosage forms that may be used to improve the treatment of critically ill patients. Critical care pharmacists should collaborate with intensivists and other experts to request the addition or deletion of a drug from the formulary.

F. Box 3 lists the elements of a drug evaluation monograph that should be reviewed before a drug is approved to the hospital formulary.

Box 3. Elements of a Drug Evaluation Monograph

Brand name	Dosage regimen recommendations
Generic name	Dosage regimen recommendations for special populations such as children, older adults, persons with renal and hepatic impairment, and persons undergoing dialysis
Manufacturer	Routes of administration and/or limitations
Therapeutic classification	Any special administration techniques (prescriber certification, smart pump limitations)
FDA status: Prescription, nonprescription, or controlled substance	Preparations available
Look-alike sound-alike names with any other FDA label-approved medications	Storage and stability considerations
Look-alike sound-alike names with any other FDA label-approved medications on the formulary	Any availability concerns (specialty pharmacy restrictions)
Date of FDA label approval	Critical review of pertinent clinical trials with salient critique and conclusions
FDA rank (priority or standard)	Critical review of comparison trials with similar or alternative agents
FDA label-approved indication	Cost analysis including annual projected costs
Unlabeled indications	Economic evaluations
Potential unlabeled uses	Pharmacoeconomic analyses available
Similar agents not on the formulary	Budget impact analysis specific to the institution
Similar agents on the formulary	Reimbursement from third-party payers
How the drug can be used when applied to available national guidelines	Are there any severe medication errors or sentinel events with this agent?
How the drug can be used when applied to hospital guidelines, protocols, or pathways	Does this medication need to be stored under specific circumstances to avoid medication errors or mix-ups?
Dosage form	Does this medication require tall man lettering labeling or precautionary or high-risk labeling to avoid potential medication errors or mix-ups?
Dosage strength	Is the manufacturer-provided labeling considered clear and safe for dispensing?
Mechanism of action	Is an abuse potential associated with the use of this medication?
Absorption	Patient education requirements
Distribution	Will this agent replace an existing agent, and should a formulary deletion take place?
Metabolism	Reason why or why not this medication should be included in the formulary Recommendation for addition to the formulary
Excretion	References
Common ADRs	
Significant or life-threatening ADRs or ADEs	
Boxed warnings	
Precautions	
Contraindications	
Drug-drug interactions	
Drug-food interactions	
Drug-laboratory test interactions	
IV incompatibilities	
IV compatibilities	
Pregnancy category	
Use during breastfeeding	

ADE = adverse drug event; ADR = adverse drug reaction; IV = intravenous(ly).

G. A comprehensive literature review should be used to determine a drug's efficacy and toxicity profile, with stronger levels of evidence guiding decisions regarding whether to add to the formulary.
   1. Prospective double-blind randomized controlled trials should have greater weight than retrospective cohort and case-control studies, though it is important to critique all types of studies regarding limitations to understand the strength of a study.
   2. Meta-analyses may also be considered, depending on the number and criteria for included studies.
   3. Case reports should be used only when no other evidence is available.

H. Pharmacists developing the drug evaluation monograph must be adept in drug literature evaluation and pharmacoeconomics.

I. Drug evaluations should contain references to evidence, and opinion statements should be noted.

J. Internal prescribing data may also be used in formulary decisions, such as
   1. Quantity of drug used over a specified time
   2. Medication use evaluation data
   3. ADR data
   4. Medication error data

K. Because trials leading to the approval of new drugs are often limited in their sample size and ability to detect ADEs (especially more long-term effects), it is prudent to observe safety profiles of new drugs before considering addition to the formulary, if possible. For example, some drugs may cause life-threatening toxicity such as hepatic failure at a rate of 1:20,000 patients, thus requiring more than 100,000 postmarketing patient exposures before generating a signal of toxicity.

L. Drugs may be added to the formulary without any use restrictions, or they may be added with restrictions, such as the following:
   1. Restriction to specific patient population meeting use criteria (e.g., level of monitoring, other therapies tried)
   2. Restriction to specific patient population at low risk of an ADE (e.g., hepatotoxicity)
   3. Restriction to specific providers

M. Once a drug is admitted for formulary approval, periodic assessments in the form of a medication use evaluation or reviews of use (including compliance with institutional guidelines), cost, safety, and efficacy should be made, preferably within 3–6 months and again in 1 year. The goal is to determine the effectiveness of the drug. Effectiveness is the use of a drug in the real-world setting outside a randomized controlled trial and differs from the efficacy reviewed before adding the drug to the formulary. Examples of approvals that may require quicker reevaluation (e.g., 3 months) are those with very limited efficacy data, concern for lack of compliance with formulary restrictions, high risk of safety events, or medications with a very high cost.

N. All drugs on the formulary within a class should be assessed annually or more often when there is an important change in prescribing information, when a landmark trial or publication affects the drug's use, or when new FDA label-approved agents are available within the drug class. Medication assessments typically prompt updates and modifications to the drug's current use.

Patient Case

Questions 6–8 pertain to the following case.

Your institution is considering adding angiotensin II (ATII) to the hospital formulary. Angiotensin II is a non-catecholamine vasopressor that is FDA approved for increasing blood pressure in distributive shock after publication of the ATHOS-3 trial. The trial, which compared intravenous ATII with placebo once patients received a certain dose of vasopressors, showed that patients who received ATII reached a hemodynamic goal more often than placebo, with no differences in mortality or safety outcomes. Your hospital formulary currently includes the catecholamine vasopressors and vasopressin. A particular concern of your institution is the cost of ATII therapy and adherence to formulary restrictions.

6. When creating the drug evaluation monograph for adding ATII, which recommendation would be most appropriate to include?
   A. Remove vasopressin from the hospital formulary to make room for ATII.
   B. Addition of statement that ATII decreases mortality in patients with distributive shock.
   C. Addition of both the intravenous and oral formulations of ATII to the formulary.
   D. Restrict use to patients who received the same dose of vasopressors as those included in the ATHOS-3 trial.

7. In considering adding ATII to the formulary, which characteristic of your institution would be most important to consider?
   A. Data from ATHOS-3 suggesting its clinical superiority to other FDA-approved medications.
   B. How often you encounter patients who require several vasopressors for management of distributive shock.
   C. The increased risk of ADEs associated with the use of ATII.
   D. Data analyses suggesting the cost-saving benefits of ATII compared with other FDA-approved medications.

8. Angiotensin II was approved by your institution last December, after presentation to the P&T committee. Which best depicts the month in which the first medication use evaluation should be completed?
   A. March of this year.
   B. June of this year.
   C. October of this year.
   D. December of this year.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: A**
Indirect costs occur from loss of employment or productivity as the result of illness (Answer A is correct). Effects from stress would be considered an intangible cost (Answer B is incorrect). Costs from fentanyl and transportation would be considered direct costs (Answers C and D are incorrect).

2. **Answer: A**
A mild ADE is defined as an ADE that resulted in a heightened need for patient monitoring with or without a change in vital signs but no ultimate patient harm, or as any adverse event that resulted in the need for increased laboratory monitoring. Enalapril-induced hyperkalemia managed by one dose of sodium zirconium cyclosilicate did not require aggressive interventions, nor did it lead to any patient harm; however, it did require increased laboratory monitoring, making Answer A correct and Answers B, C, and D incorrect.

3. **Answer: C**
A severe ADE results in patient harm, prolonged hospitalization, transfer to a higher level of care, permanent organ damage, or death, which did occur in this case (Answer D is incorrect). Haloperidol-induced torsades de pointes is a life-threatening dysrhythmia (end-organ damage) that required aggressive and successful management with intravenous magnesium and patient transfer to a higher level of care (Answer C is correct). A moderate ADE is defined as an ADE that resulted in the need for aggressive intervention with antidotes or an increased length of hospital stay (Answer B is incorrect). A mild ADE is defined as an ADE that resulted in a heightened need for patient monitoring with or without a change in vital signs but no ultimate patient harm, or as any adverse event that resulted in the need for increased laboratory monitoring, which did not occur in this case, making Answer A incorrect.

4. **Answer: A**
Because this patient developed lisinopril-induced angioedema, had no history of allergy to angiotensin-converting enzyme inhibitors, and missed no lisinopril doses, his ADE was not a preventable error (Answer B is incorrect). This was also not a medication error, given the information provided (Answer C is incorrect). There is no such thing as a preventable ADR because an ADR occurs from normal use of a medication and is therefore by definition nonpreventable (Answer D is incorrect). Therefore, this would be considered an ADE (Answer A is correct).

5. **Answer: C**
A severe ADE is defined as an ADE that results in patient harm, prolonged hospitalization, transfer to a higher level of care, permanent organ damage, or death, with the probable ADE causality nomogram score. Because this patient developed life-threatening lisinopril-induced angioedema and needed a tracheotomy, resulting in patient harm, hospitalization, and transfer to a higher level of care, this case meets the criteria for a "severe" ADE, making Answer C correct and Answers A, B, and D incorrect.

6. **Answer: D**
When considering what to include in a drug evaluation monograph for formulary addition, it is important to consider the data analyses available supporting its use. According to what was provided in the case, ATII was only compared with placebo (Answer A is incorrect), did not show any significant differences in mortality (Answer B is incorrect), and was studied in an intravenous formulation (Answer C is incorrect). Answer D is correct because considering the inclusion criteria for the study would be helpful for formulary restriction, especially given the institutional concern of cost.

7. **Answer: B**
Considerations for formulary addition are important before P&T committee proposals. According to the ATHOS-3 trial, ATII was not compared with other currently available medications (Answer A is incorrect), there were no significant differences in ADEs (Answer C is incorrect), and no data analyses are available suggesting that ATII is cost-saving (Answer D is incorrect). Considering how often you might require the use of ATII would be an important initial consideration (Answer B is correct).

8. **Answer: A**
The first medication use evaluation is recommended to occur within 3–6 months of formulary addition. Therefore, March would be the most appropriate month (Answer A is correct). Although June would fall in the

3- to 6-month range, given the novelty of this agent and its cost, it would be important to conduct a medication use evaluation as soon as possible (Answer B is incorrect). October and December would be too long to wait for an evaluation to be performed (Answers C and D are incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: B**
The cost of a medication would be considered a variable medical cost (Answer B is correct). The cost of the lights in the clean room would be considered a fixed medical cost (Answer A is incorrect). The time resulting in loss of work for a patient requiring this medication would be considered an indirect cost (Answer C is incorrect). Additional use of blood products after administration of this medication would be considered an incremental cost (Answer D is incorrect).

2. **Answer: B**
Given the need to consider efficacy and cost, a cost-effectiveness analysis would be the best type of analysis (Answer B is correct). A cost-benefit analysis would be used if your institution were purely interested in a monetary outcome (Answer A is incorrect). A cost-minimization analysis would only evaluate the difference in costs between these medications (Answer C is incorrect). A cost-utility analysis includes the patient perspective, which was not a focus of this evaluation (Answer D is incorrect).

3. **Answer: D**
A preventable ADE, by definition, is a medication error that occurs and reaches the patient to cause harm because of a breach of standard professional behavior or practice. The patient had a documented history of cefuroxime allergy with shortness of breath but still received ceftriaxone and developed another life-threatening anaphylactic reaction; this is a medication error, and the anaphylactic reaction is the harm. Allergy cross-reactivity between cefuroxime and ceftriaxone is well documented for patients with a history of shortness of breath because of similarity in side chains; therefore, this is a preventable ADE (Answer D is correct). Although this case of ceftriaxone-induced life-threatening anaphylaxis is an ADE, a preventable ADE best describes this case (Answer C is incorrect). In general, *ADRs* and *side effects* are synonymous terms, and a ceftriaxone-induced allergy is an ADR; however, a preventable ADE best describes this case (Answers A and B are incorrect).

4. **Answer: D**
Prothrombin complex concentrate would be the best tracer for an ADE associated with apixaban (Answer D is correct). Coagulation factor VIIa would not be effective for this indication (Answer A is incorrect). Phytonadione would be an appropriate tracer for a warfarin-induced bleeding event (Answer B is incorrect). Protamine would be an appropriate tracer for a heparin- or enoxaparin-induced bleeding event (Answer C is incorrect).

5. **Answer: C**
The guidelines provide several recommendations, with only one having a strong recommendation, which is based on the dispensing of medications and streamlining them as only being from the pharmacy (Answer C is correct). The other potential answers are potential opportunities to improve medication safety but are based on weak recommendations (Answers A, B, and D are incorrect).

6. **Answer: B**
The guidelines suggest (not recommend) that patient/caregiver-reported outcomes be integrated into routine patient care (Answer A is incorrect). Data analyses on the benefit of these interactions are limited to non-ICU environments, though the theoretical benefits listed in Answers C and D may be true (Answers C and D are incorrect; Answer B is correct).

7. **Answer: B**
To determine the probability of a drug interaction between meropenem and valproic acid, the drug interaction probability scale should be used (Answer B is correct). A drug interaction database would provide general information and is not a patient-specific evaluation (Answer A is incorrect). The Naranjo nomogram investigates the likelihood of an ADE but is not specific to a drug interaction (Answer C is incorrect). The RUCAM evaluates the likelihood of drug-induced liver injury (Answer D is incorrect).

8. **Answer: C**
Broad-spectrum antimicrobials should be restricted to the appropriate clinicians with expertise in this area (Answer C is correct). Restricting use to intensivists would not be appropriate because non-ICU patients may require this agent (Answer A is incorrect). Although FDA approved only for complicated UTIs, restriction to pneumonia would not be appropriate (Answer B is incorrect). Cardiac monitoring is not required for this medication (Answer D is incorrect).

Appendix 1. Adverse Drug Event Reporting Form

ADVERSE EVENT INFORMATION ADE#:

1. NAME	2. PATIENT ID #	3. LOCATION	4. AGE	5. SEX	6. REACTION ONSET DATE
					7. DATE OF REPORT

8. DESCRIBE REACTION AND ITS MANAGEMENT. (Continue on the back if necessary. Use Arial Narrow Font Size 10)

9. Check all appropriate

	Patient died
	Reaction treated with drug
	Resulted in or prolonged inpatient hospitalization
	None of the above

10. Did event abate after discontinuing the drug?

	YES
	NO
	MAYBE

11. Was patient's electronic allergy/ ADE profile updated?

	YES
	NO

(If no, please explain on second page)

12. RELEVANT TESTS/LABORATORY DATA

SUSPECTED DRUG(S) INFORMATION

13. SUSPECTED DRUG(S) Give manufacturer & lot number for vaccine/ biologics/ biotechnological

17. DATES OF ADMINISTRATION

14. DOSE AND FREQUENCY	15. ROUTE OF ADMINISTRATION	18. DURATION OF ADMINISTRATION

16. INDICATION(S) FOR USE

CONCOMITANT DRUG HISTORY

19. CONCOMITANT DRUGS AND DATES OF ADMINISTRATION (Exclude those used to treat the reaction)

20. OTHER RELEVANT HISTORY (e.g., diagnoses, medical history, allergies, pregnancy)

INITIAL REPORTER (In confidence)

JCAHO Standard PI. 2.20 states that all serious adverse drug reactions are intensely analyzed. Standard MM. 6.20 maintains that the responsible individual complies with internal and external reporting requirements for adverse drug reactions. (2006 Comprehensive Accreditation Manual for Hospitals)

Please take the time to complete this form for each suspected adverse drug reaction, and forward it to the Department of Pharmacy for reporting at the next Adverse Drug Reaction Subcommittee meeting.

Date	Time	MD Notified about Possible ADR	Pharmacist's Signature

Submission of a report does not necessarily constitute an admission that the drug caused the reaction

NAME AND ADDRESS OF REPORTER (Including zip code)

TELEPHONE NO. (Include area code)

HAVE YOU ALSO REPORTED THIS REACTION TO THE MANUFACTURER?

	YES
	NO

Appendix 1. Adverse Drug Event Reporting Form *(continued)*

KJMC ADVERSE DRUG EVENT REPORTING FORM
Modified Naranjo Criteria for Causality (also called ADR Probability Scale)

ASSESSMENT	YES	NO	DON'T KNOW (not conducted, not reported)	SCORING SYSTEM
1. Are there previous reports (in the literature) of this reaction? (If no, please provide documentation of search strategy)	+1	0	0	Based on the total score, circle the term that best defines this ADR: ≥ 9 Definite 5–8 Probable 1–4 Possible ≤ 0 Doubtful
2. Did the ADR appear after the suspected drug was administered, confirming a temporal relationship? (If no, please explain)	+2	-1	0	
3. Did the ADR improve when the drug was discontinued or a specific antagonist was administered? (If no, please explain)	+1	0	0	
4. Did the ADR reappear when the drug was readministered?	+2	-1	0	
5. Are there alternative causes (other than the drug) that could have caused the ADR? (If yes, please explain)	-1	+2	0	
6. Did the ADR reappear when a placebo was given?	-1	+1	0	
7. Was the drug detected in the blood or other fluids in toxic concentrations?	+1	0	0	
8. Evaluate the dose-response relationship; was the ADR more severe when the dose was increased or less severe when the dose was decreased?	+1	0	0	
9. Did the patients have a similar reaction to the same or similar drugs (i.e., same drug class) in a previous exposure?	+1	0	0	
10. Consider signs, symptoms, and laboratory values other than drug concentrations. Was the ADR confirmed by objective evidence?	+1	0	0	
TOTAL SCORE				

Source: Naranjo CA, Busto U, Sellers EM, et al. A method for estimating the probability of adverse drug reactions. Clin Pharmacol Ther 1981;30:239-45.

Appendix 2. Drug Interaction Probability Scale

The drug interaction probability scale (DIPS) is designed to assess the probability of a causal relationship between a potential drug interaction and an event. It is patterned after the Naranjo ADR probability scale (Naranjo CA, Busto U, Sellers EM, et al. A method for estimating the probability of adverse drug reactions. Clin Pharmacol Ther 1981;30:239-45).

Directions

- Circle the appropriate answer for each question and add up the total score.
- Object drug = Drug affected by the interaction.
- Precipitant drug = Drug that causes the interaction.
- Use the Unknown (Unk) or Not Applicable (NA) category if (a) you do not have the information or (b) the question is not applicable (e.g., no dechallenge; dose not changed).

Questions	Yes	No	NA/Unk
1. Are there previous credible reports of this interaction in humans reported in the literature?	+1	–1	0
2. Is the observed interaction consistent with the known interactive properties of the precipitant drug?	+1	–1	0
3. Is the observed interaction consistent with the known interactive properties of object drug?	+1	–1	0
4. Is the event consistent with the known or reasonable time course of the interaction (onset or offset)?	+1	–1	0
5. Did the interaction remit upon dechallenge of the precipitant drug with no change in the object drug? (if no dechallenge, use Unknown or NA and skip question 6)	+1	–2	0
6. Did the interaction reappear when the precipitant drug was readministered in the presence of continued use of object drug?	+2	–1	0
7. Are there reasonable alternative causes for the event?[a]	–1	+1	0
8. Was the object drug detected in the blood or other fluids in concentrations consistent with the proposed interaction?	+1	0	0
9. Was the drug interaction confirmed by any objective evidence consistent with the effects on the object drug (other than drug concentrations from question 8)?	+1	0	0
10. Was the interaction greater when the precipitant drug dose was increased or less when the precipitant drug dose was decreased?	+1	–1	0

[a]Consider clinical conditions, other interacting drugs, lack of adherence, risk factors (e.g., age, inappropriate doses of object drug). A NO answer presumes that enough information was presented so that one would expect any alternative causes to be mentioned. When in doubt, use Unknown or NA designation.

Total Score _____ Highly Probable: > 8
Probable: 5–8
Possible: 2–4
Doubtful: < 2